



Off by One

Ontario Math C3.2 — read, find, and fix code

Name: _____

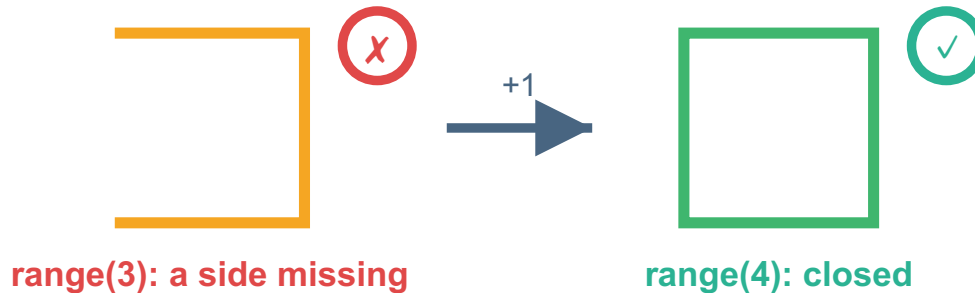
Date: _____

★ I can fix a loop whose repeat number is one too many or too few.

One too many or one too few

A very common bug is being OFF BY ONE — the repeat number is just one more or one less than it should be.

In a turtle shape, off by one is easy to see: a square needs `range(4)`; `range(3)` leaves a side missing.



`range(3)` leaves the square open; `range(4)` closes it.

1 Too many

Bit should repeat 5 times, but the code says repeat 6. How many too many is that, and what should it be? _____

2 Too few

Bit should repeat 6 times, but the code says repeat 5. Is it too many or too few? What should it be?

3 Fix the turtle

This should draw a square: `for i in range(5): bit.forward(50); bit.right(90)`. The range is off by one. What should `range()` be?

4 Challenge

Bit's square code is `for i in range(3): bit.forward(50); bit.right(90)`. What does Bit draw now, and what one change fixes it? Explain why.

Big idea

Off-by-one is a tiny bug with a big effect. Always check the repeat number against the goal — one more or one less changes the whole result.



Off by One

Quick facilitation notes & answer key

What this worksheet teaches

Students fix off-by-one loop bugs — a loop that repeats one too many or one too few — in both block (count) and shape (a square that does not close) cases. The method here is SHOULD → DOES → FIX: say what the code SHOULD do, run it to see what it DOES, then FIX the one thing that is wrong. No coding experience needed.

Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

How to lead it (about 30 minutes)

1. Show it (you do). A repeat that is one off makes Bit do one too many or one too few; for a shape, it leaves a side open.
2. Do one together. Exercise 1 — repeat 6 is one too many; fix to repeat 5.
3. Let them work. Students fix a too-few count (Ex 2), a square missing a side (Ex 3), and an open triangle-shape (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — one too many; repeat 5. Exercise 2 — too few; repeat 6.

Exercise 3 — repeat 4 times (a square has 4 sides). Exercise 4 (challenge) — only 3 sides are drawn (an open shape); change the repeat to 4 to close the square.

If students get stuck — and ways to adjust

Watch for: fixing by a big jump. Off-by-one means change the count by just one.

Make it easier: count the sides the shape needs versus the sides it draws.

Make it harder: ask how an off-by-one looks different for a count versus a shape.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

You'll know it worked when

The child fixes the repeat by one so the count is right or the square closes.